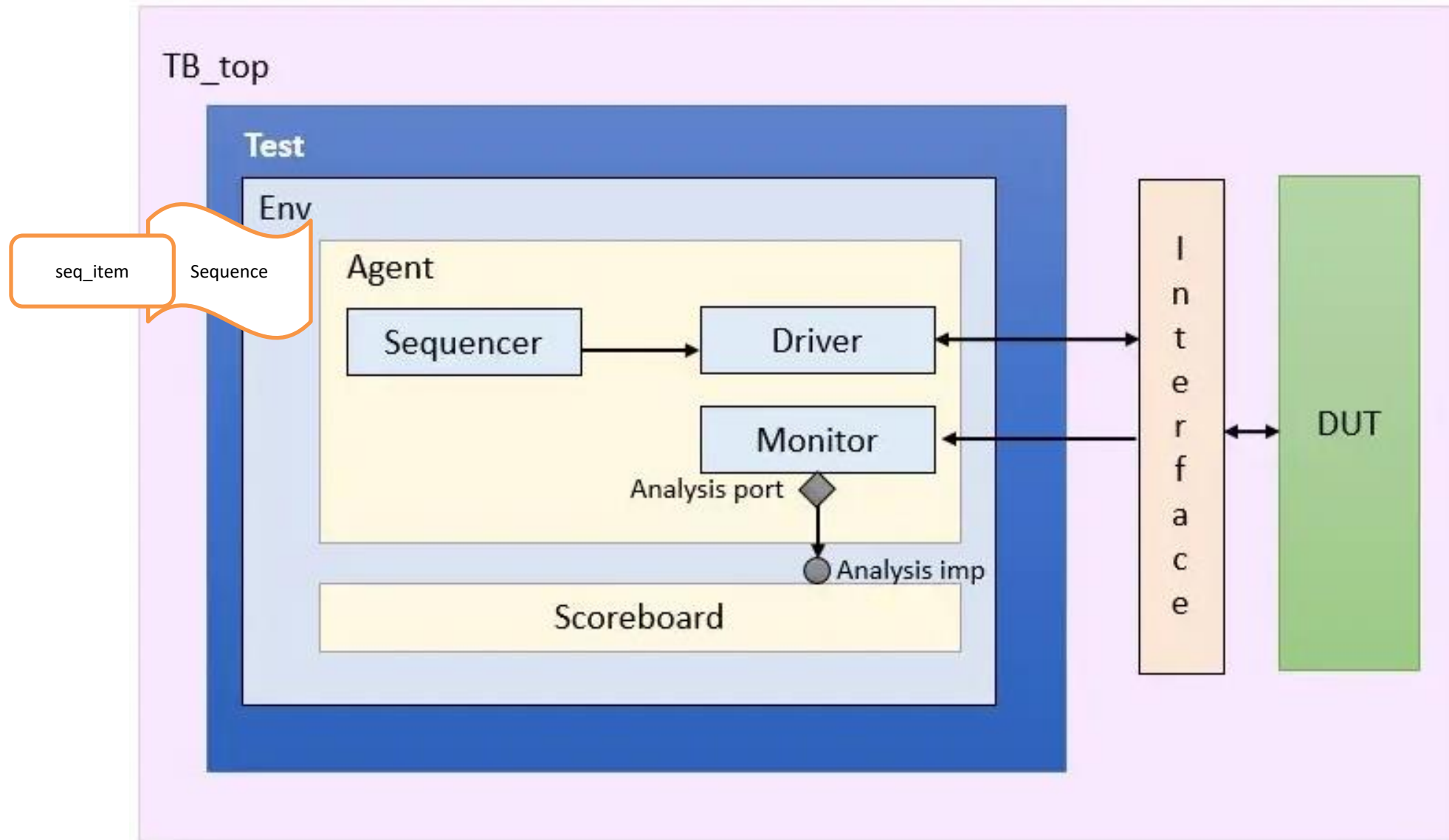


Functional Verification Flow

Yasir Farooq

Recall the UVM based Testbench



Functional Verification Flow

Tools:

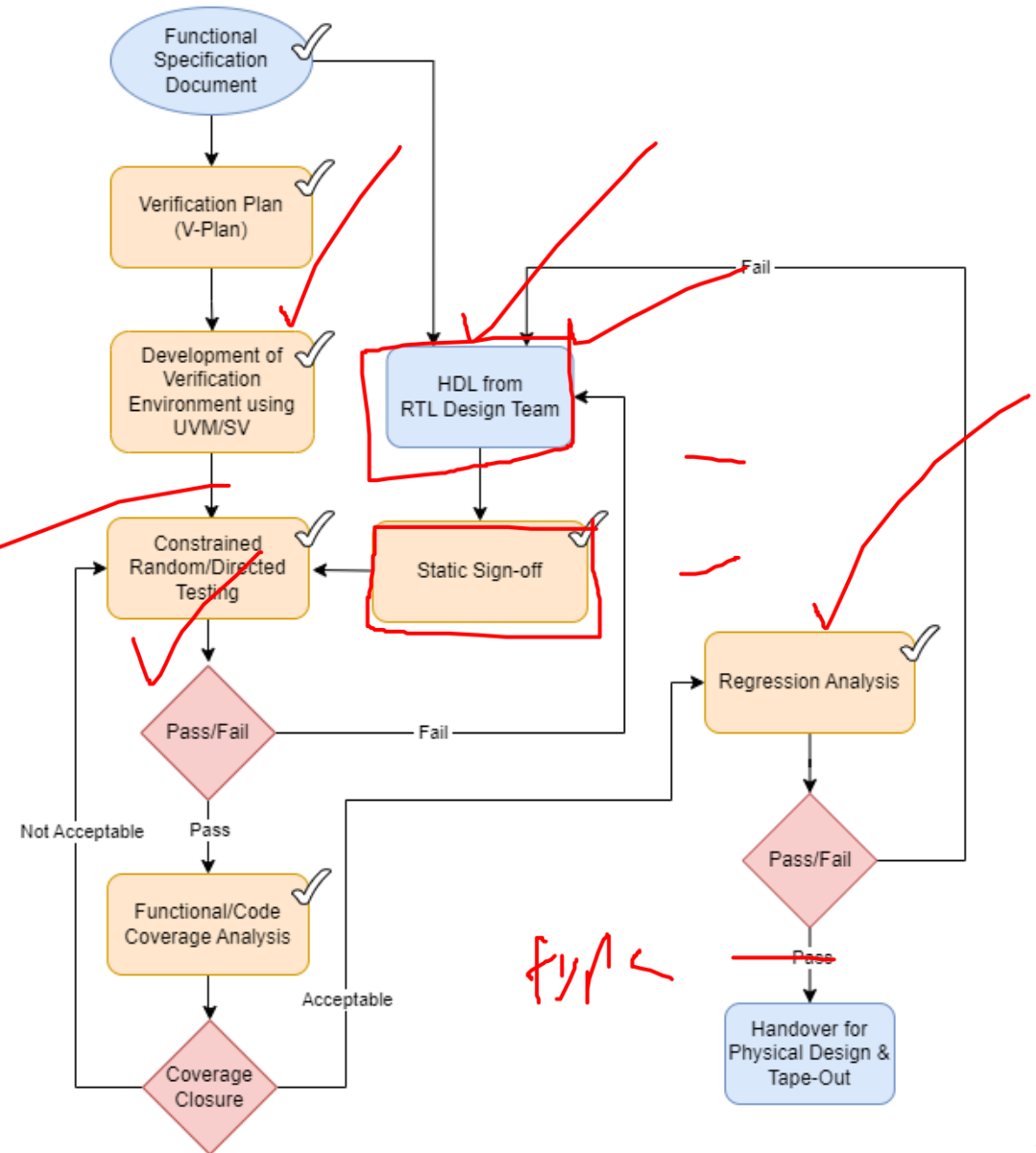
Synopsys SpyGlass ✓
Synopsys VCS ✓

Programming and Scripting Languages:

SystemVerilog
C/C++
Python
Shell
TCL

Functional Verification Methodology:

Universal Verification Methodology (UVM)

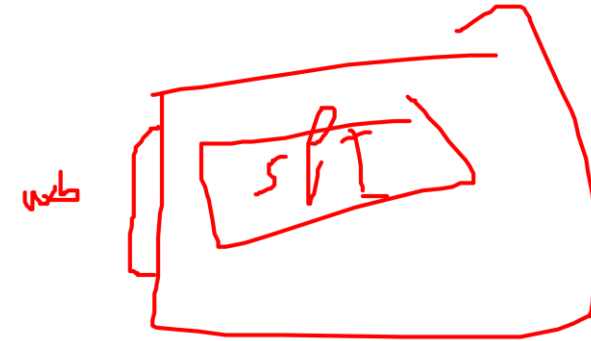
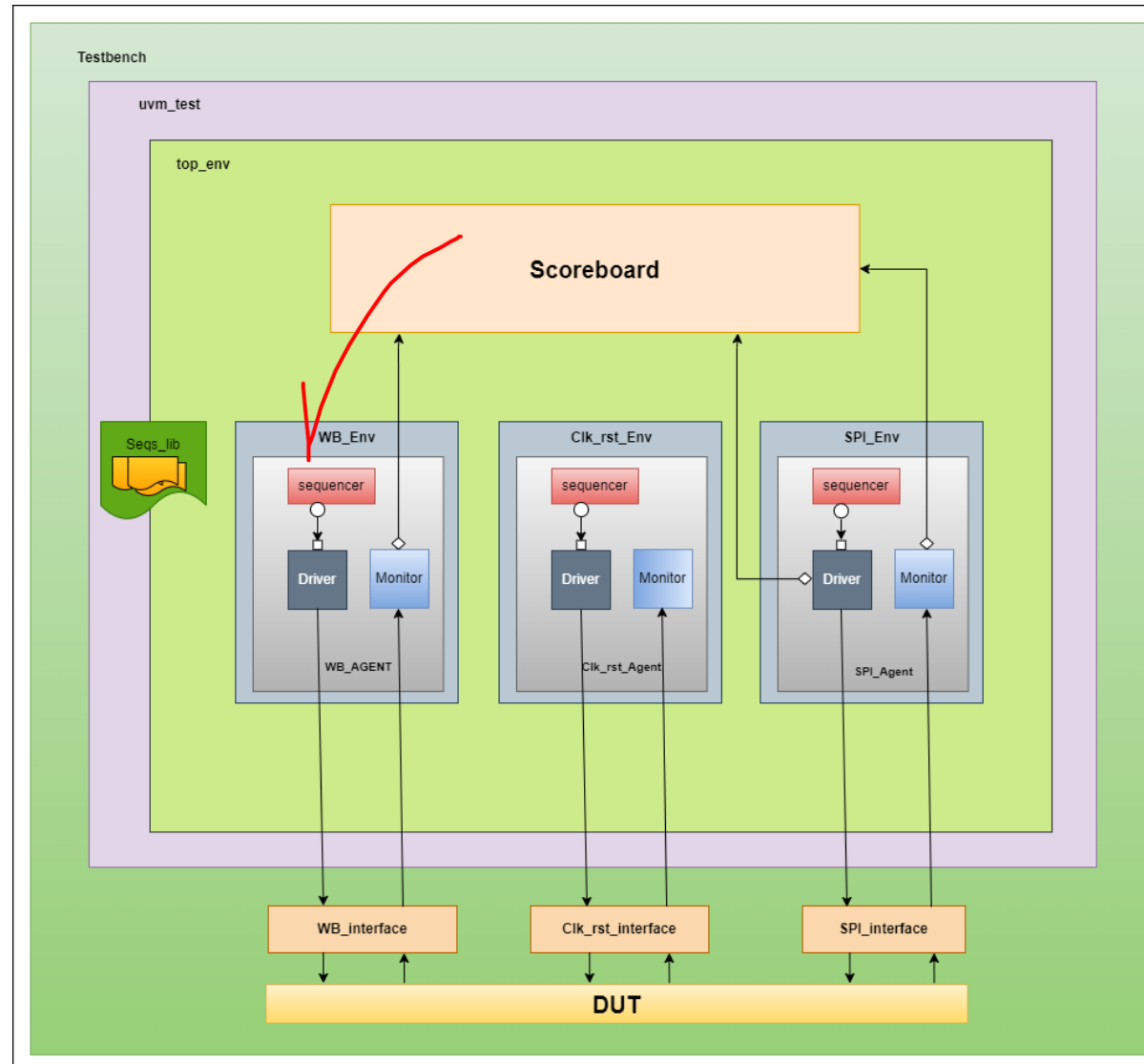


Design Verification Plan



SPI Master DV Plan				
Serial No.	Test Name	Feature	Test Description	Result
1	Reset Test	Reset	Apply reset and check the default reset values of config	Pass
2	CSRs RW Test	Read/Write CSRs	Write 8 bit value in the register and then read it. SPSR register has [7:6] bits which can be WR, other [5:0] are read only	Pass
3	Mode 0 of SPI Transfer (CPOL=0, CPHASE=0)	SPI Modes WRT Polarity and Clock Phase	Data is sampled on rising edge and shifted out on the falling edge. Clock Polarity is at logic low in Idle	Pass
4	Random CPOL and CPHASE Test	Coverage enhancement	Run 128 iteration by randomizing CPOL & CPHASE field of spcr[3:2]	Pass

Development of Verification Environment



Coverage Closure

- ▶ **Code Coverage**

- Line
- Toggle
- FSM
- Condition

- ▶ **Functional Coverage**

- CoverGroups
 - Coverpoints



Regression Analysis and Functional Signoff

- ▶ All the developed tests are ran using regression analysis
- ▶ After correcting all bugs and coverage closure
- ▶ The HDL design is freeze after passing regression analysis
- ▶ The verification report is shared with the HDL designers for functional signoff
- ▶ The verified IP can be used as soft IP for the customized projects



UVM based Verification Projects

Project 1 Functional Verification of Wishbone Compliant Memory

Project 2 Functional Verification of APB Compliant SRAM